MAS 2001: Solution Scheme

B. Tech. III (CSE|AIML|ECE|EE)

Section - A

Q. A (1) 6 (2)

QA (2) B (2)

QA (3) (C) D (1+1)

Section (B)

Q. B (1) Let $X \sim P(\lambda_1)$ and $Y \sim P(\lambda_2)$

Thus, we have,

$$P(X=x) = \frac{e^{-\lambda_1} \lambda_1^{x}}{x!} \quad ; \quad x = 0,1,2,\ldots, \ \lambda_1 > 0$$

$$P(Y=y) = \frac{e^{-\lambda_2} \lambda_2^{x}}{x!} \quad ; \quad x = 0,1,2,\ldots, \ \lambda_2 > 0$$

(1)

Using given condition, we get (1)

$$\lambda_1 e^{-\lambda_1} = \frac{\lambda_1^2 e^{-\lambda_1}}{2!} \quad \text{and} \quad \frac{\lambda_2^2 e^{-\lambda_2}}{2!} = \frac{\lambda_2^3 e^{-\lambda_2}}{3!}$$

Solving (1), we get $\lambda_1 = 2$ and $\lambda_2 = 3$ as $\lambda_1, \lambda_2 > 0$

Now $V(X) = \lambda_1 = 2$; $V(Y) = \lambda_2 = 3$ (1)

Hence
$$V(X-2Y) = 1^2 V(X) + (-2)^2 V(Y)$$
$$= 2 + 4 \times 3 = 14 \text{ Ans}$$
(1)

3.B2: Let time of speaking is the exponentially distributed r.v. (X) with parameter $\lambda = \frac{1}{4}$.

(i) $P(X>6) = \int_6^\infty \frac{1}{4} e^{-\frac{1}{4}x} dx = \frac{1}{4}\int_6^\infty e^{-x/4} dx$

$= \frac{1}{4} \frac{e^{-x/4}}{-1/4}\Big|_6^\infty = 0.2231$ (1)

(ii) $P(7<X<12) = \int_7^{12} \frac{1}{4} e^{-x/4} dx = \frac{1}{4} \cdot \frac{e^{-x/4}}{-1/4}\Big|_7^{12}$

$= -e^{-x/4}\Big|_7^{12} = -\left[e^{-3} - e^{-7/4}\right]$ (1)

$= e^{-7/4} - e^{-3} = 0.17377 - 0.04978 = 0.12399$

(iii) $P(X \leq 5) = \int_0^5 \frac{1}{4} e^{-x/4} dx = \cancel{\frac{1}{4}} \frac{e^{-x/4}}{-\cancel{1/4}}\Big|_0^5$ (1)

$= -\left[e^{-5/4} - e^0\right] = 1 - e^{-5/4}$

$= 1 - 0.2865 = 0.7135$

(iv) Mean $= \frac{1}{\lambda} = \frac{1}{1/4} = 4$; Variance $= \frac{1}{\lambda^2} = \frac{1}{(1/4)^2} = 16$

(1/2) (1/2)

QB(3) since $f(x) = \frac{1}{2}$; $-1 \le x \le 1$

So, we have $\mu = E(X) = 0$

$\sigma^2 = V(X) = \frac{1}{3}$ (1)

Also, from chebyshev's inequality, we have

$$P[|X-\mu| \geq K\sigma] \leq \frac{\sigma^2}{K^2\sigma^2} = \frac{1}{K^2}$$

$$\Rightarrow P[|X-\mu| \geq 2\sigma] \leq \frac{\sigma^2}{4\sigma^2} = \frac{1}{4} \quad (1)$$

- Thus the upper bound of the given probability is $\frac{1}{4}$.

Now from the values of μ and σ^2, we have

$$P\left[|X-0| \geq \frac{2}{\sqrt{3}}\right] = 1 - P\left[|X| \leq \frac{2}{\sqrt{3}}\right] \quad (2)$$

$$= 1 - P\left[-\frac{2}{\sqrt{3}} \leq X \leq \frac{2}{\sqrt{3}}\right] = 1 - P[-1.1547 \leq X \leq 1.1547]$$

$$= 1 - P[-1 \leq X \leq 1] = 1 - 1 = 0$$

Q B4:- Let X be the random variable representing the number of minutes past 9 that the passenger arrives at station. So, $f(x) = \frac{1}{30}$ (1)

(a) He has to wait for less than 6 minutes if he arrives between 9:09 and 9:15 or 9:24 and 9:3[illegible]

So required Probability =

$$P(9 < X < 15) + P(24 < X < 30) \quad (1.5)$$

$$= \int_9^{15} \frac{1}{30}\,dx + \int_{24}^{30} \frac{1}{30}\,dx = \frac{1}{5} + \frac{1}{5} = \frac{2}{5}$$

(b) Required Probability $= P(0 < X < 5) + P(15 < X < 20)$

$$= \frac{1}{6} + \frac{1}{6} = \frac{1}{3} \quad (1.5)$$

Section: C

Q.C (a) Given $X \sim N(2.6, 34.5)$, $\sigma = 5.87$

(i) We need $P(X > 6)$

$$Z = \frac{X - \mu}{\sigma} = \frac{6 - 2.6}{5.87} = 0.58 \quad (1)$$

$$P(X > 6) = P(Z > 0.58) = 1 - P(Z < 0.58) = 0.281$$

(ii) $P(X < 1)$

$$Z = \frac{X - \mu}{\sigma} = \frac{1 - 2.6}{5.87} = -0.27 \quad (1)$$

$$P(X < 1) = P(Z < -0.27) = 0.3936$$

(iii) $P(1.5 < X < 4.6)$

$z_1 = \frac{x_1 - \mu}{\sigma} = -0.19$; $z_2 = \frac{x_2 - \mu}{\sigma} = 0.34$

$P(1.5 < X < 4.6) = P(-0.19 < Z < 0.34) = P(Z < 0.34) - P(Z < -0.19)$

$= 0.6331 - 0.4247 = 0.2084$ (1)

(iv) Let Y be the randomly selected days in which the rainfalls range from 1.5 mm to 4.6 mm. The probability of a day having such amount of rainfall is $p = 0.2084$

There $q = 1 - p = 0.7916$. we have

$P(Y = y) = \binom{7}{y} (0.2084)^y (0.7916)^{7-y}$; $y = 0, 1, 2, \ldots, 7$

$\therefore P(Y \le 2) = P(Y=0) + P(Y=1) + P(Y=2)$ (1)

$= \binom{7}{0} (0.7916)^7 + \binom{7}{1} (0.2084)(0.7916)^6 + \binom{7}{2} (0.2084)^2 (0.7916)^5 \approx 0.83722$.

Q.C(b) Let n and p be the parameters. Thus we have,

$P(X = x) = \binom{n}{x} p^x q^{n-x}$ where $x = 0, 1, \ldots, n$ and $p + q = 1$

By question, we have (1)

mean $= np = 5/3$; $P(X=2) = P(X=1)$

${}^nC_2 \, p^2 q^{n-2} = {}^nC_1 \, p^1 q^{n-1} \Rightarrow p = \frac{1}{3}, q = \frac{2}{3}$

we have $n=5$

Now, the Variance of the distribution $= npq = 5 \times \frac{1}{3} \times$

$= \frac{10}{9}$

$P(X = \text{at least } 1) = P(X \geq 1) = 1 - P(X \leq 0) = 1 - P(X=0)$

$= 1 - {}^5C_0 \left(\frac{1}{3}\right)^0 \left(\frac{2}{3}\right)^5 = \frac{211}{243}$ (1.5)

$P(X = \text{at most } 1) = P(X=0) + P(X=1)$

$= \left(\frac{2}{3}\right)^5 + {}^5C_1 \left(\frac{1}{3}\right)^1 \left(\frac{2}{3}\right)^4 = \frac{112}{243}$

(1.5)